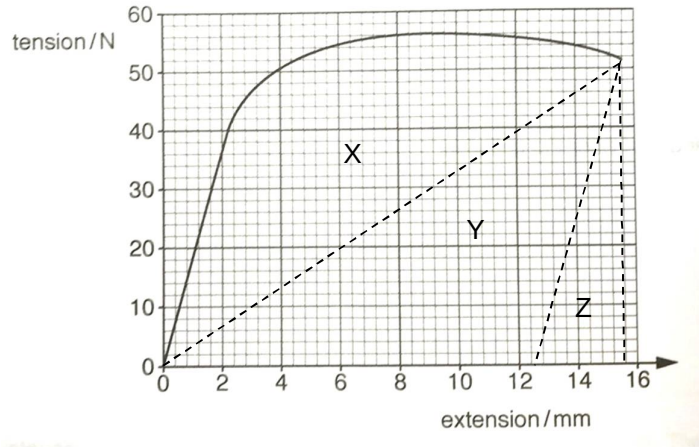


- 8 The tension in a sample of wire varies with extension as shown in the diagram below.



The graph shows that the wire undergoes two types of deformation as it is extended to 15.6 mm. In the region where Hooke's law applies, the deformation is elastic and the wire will lose this deformation when the tension is released. In the region where Hooke's law does not apply, the deformation is plastic and the wire retains this deformation when the tension is released.

Which area represents the elastic potential energy that is stored in the wire when it is extended to 15.6 mm?

- A Area Z
- B Area X + Area Y
- C Area Y + Area Z
- D Area X + Area Y + Area Z

Ans : A

0 Air in a bicycle pump is forced through a valve at a constant pressure P_0 to supply air to a